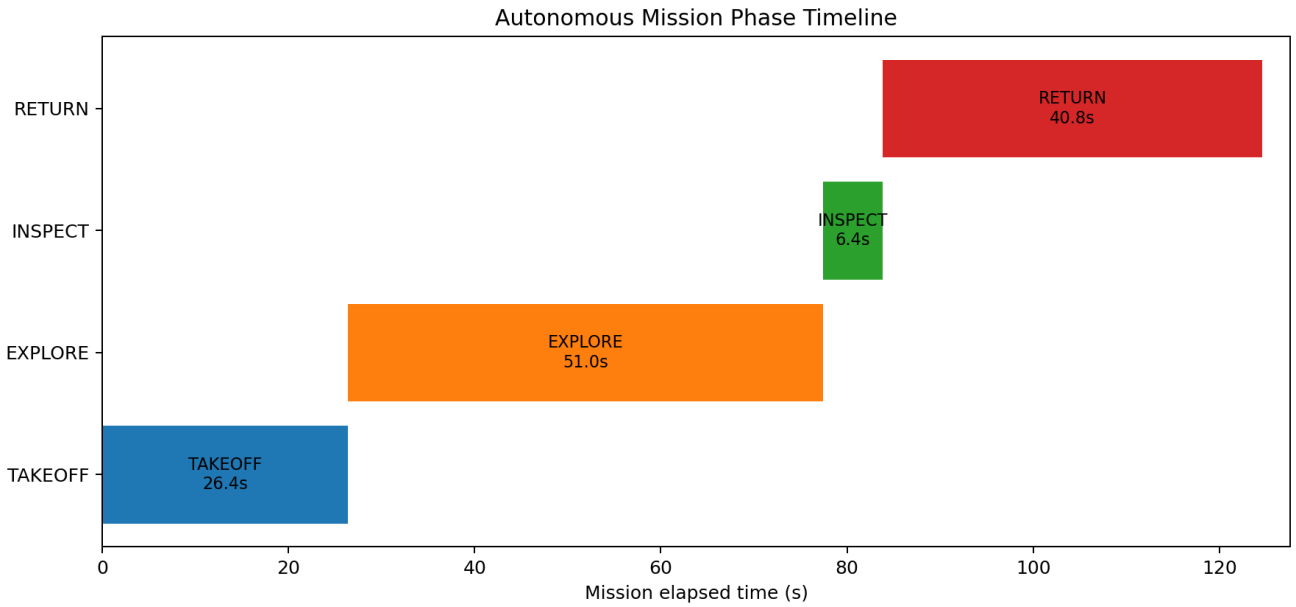
ROS 2 Autonomous Search-and-Rescue Drone

Runtime Validation Report - PX4 SITL / Gazebo Harmonic / ROS 2 Humble

This report summarizes measured evidence from the 19 Oct 2026 end-to-end simulation run. The run demonstrated autonomous OFFBOARD takeoff, frontier-based exploration, A* path generation, YOLO person detection with RGB-D 3-D localization, inspection hold, return-to-launch, landing/disarm, and mission completion. The project is simulation-first and is not a flight-certified system.

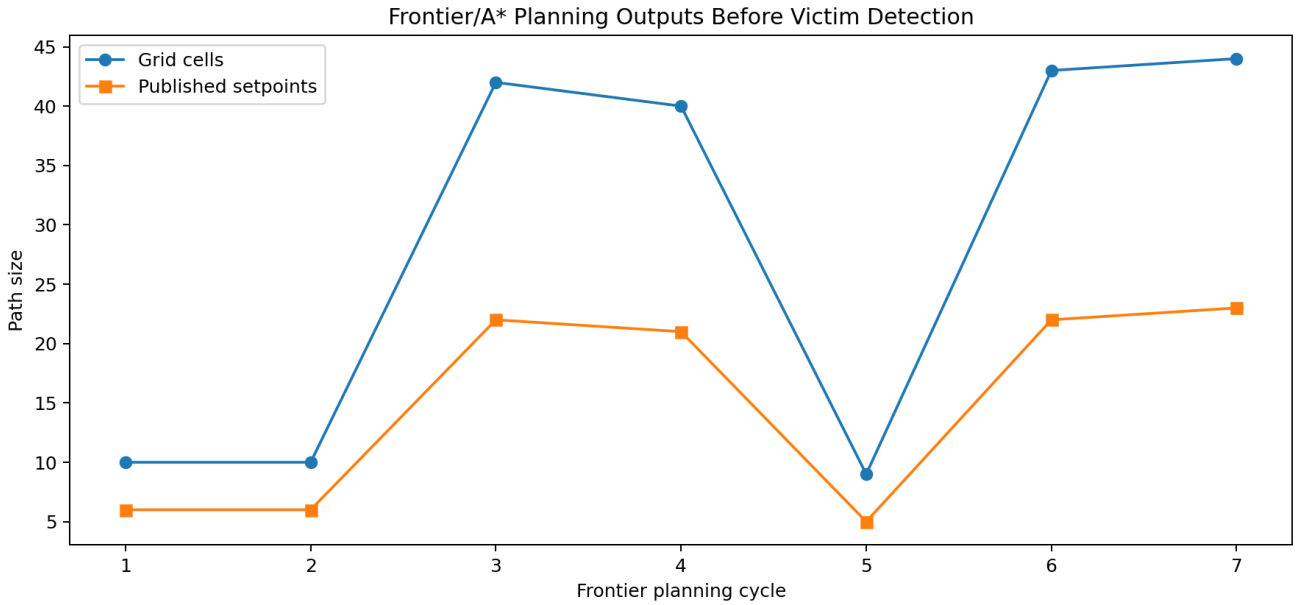
Measured item	Result
Mission completion	124.60 s elapsed
State sequence	TAKEOFF -> EXPLORE -> INSPECT -> RETURN -> COMPLETE
Unique victim candidates	1
Frontier plans logged	7
Frontier path range	9-44 grid cells
Latest YOLO confidence	0.883-0.908
Latest RGB-D range	7.976-7.985 m
Raw demo recording	107.7 s, 1920x1080

Mission timeline



Autonomous exploration and planning evidence

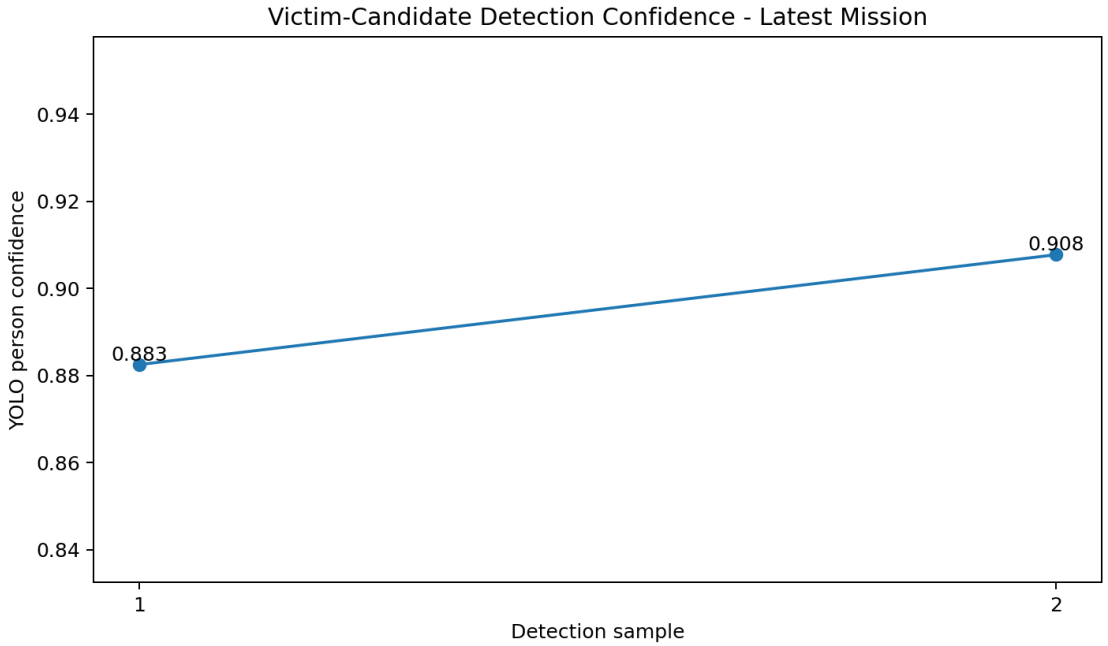
The frontier explorer generated multiple candidate paths before the victim was detected. The recorded run contains seven logged planning cycles. The path length varied as the projected OctoMap changed during flight; these values are runtime observations, not synthetic benchmark data.



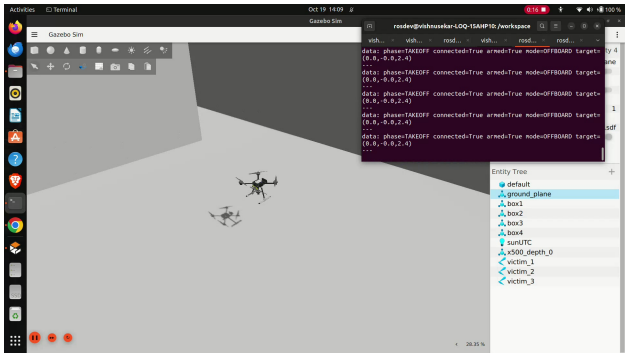
Plan	Grid cells	Setpoints	Goal cell
1	10	6	(9, 31)
2	10	6	(9, 31)
3	42	22	(24, 69)
4	40	21	(23, 72)
5	9	5	(3, 30)
6	43	22	(21, 11)
7	44	23	(23, 8)

Victim-candidate detection and 3-D localization

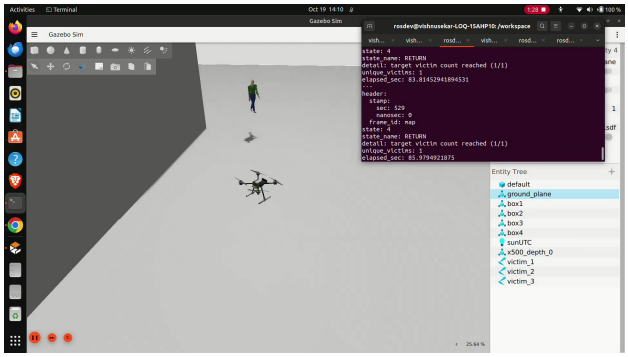
The latest mission produced person detections associated with victim_id 1. The confidence values below are YOLO model confidences and must not be interpreted as detection accuracy. Depth and TF were used by the runtime node to publish positions in the map frame.



Sample	Confidence	Distance (m)	Map position (m)
1	0.883	7.976	(-7.28, -3.21, 1.85)
2	0.908	7.985	(-7.84, -3.54, 1.82)



Recorded simulation: autonomous flight



Recorded simulation: human model visible during victim-search phase

Engineering interpretation

Validated in simulation: PX4/MAVROS connectivity, OFFBOARD arming, takeoff and altitude hold, depth-derived mapping, frontier exploration, A* path output, path following, person detection, RGB-D 3-D localization, victim deduplication, inspection hold, RTL, disarm, and COMPLETE mission state.

Known runtime warnings: MAVROS time-sync resets and repeated "PositionTargetGlobal failed because no origin" messages were observed during RTL. The mission nevertheless completed and the vehicle disarmed. These warnings should be retained as documented limitations rather than hidden.

Scope: this evidence supports an end-to-end autonomous SAR simulation claim. It does not establish real-world flight safety, medical victim classification accuracy, or statistically meaningful detection/mission success rates. Multiple repeated trials would be required for those claims.